

LN1.1

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1 Vlastnosti obecného skalárního součinu

Pro každý skalární součin platí:

$$\begin{aligned}\langle \mathbf{u}, \mathbf{v} \rangle &= \overline{\langle \mathbf{v}, \mathbf{u} \rangle} \\ \langle \mathbf{u}, t\mathbf{v} \rangle &= t\langle \mathbf{u}, \mathbf{v} \rangle \\ \langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle &= \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle \\ \langle t\mathbf{u}, \mathbf{v} \rangle &= \bar{t}\langle \mathbf{u}, \mathbf{v} \rangle \\ \langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle &= \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle \\ \|\mathbf{u} + \mathbf{v}\|^2 &= \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle\end{aligned}$$

Všechny tyto rovnosti jsou vysvětlené a dokázané ve skriptech.

a)

$$\begin{aligned}\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2 &= \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle - \langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle + \\ &+ \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle - \langle \mathbf{u}, \mathbf{u} \rangle - \langle \mathbf{u}, -\mathbf{v} \rangle - \langle -\mathbf{v}, \mathbf{u} \rangle - \langle -\mathbf{v}, -\mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \overline{\langle \mathbf{u}, \mathbf{v} \rangle} + \\ &+ \langle \mathbf{u}, \mathbf{v} \rangle - (-1)\overline{\langle \mathbf{u}, \mathbf{v} \rangle} + \langle \mathbf{v}, \mathbf{v} \rangle + (-1)\overline{\langle \mathbf{u}, \mathbf{v} \rangle} = 2\langle \mathbf{u}, \mathbf{v} \rangle + 2\overline{\langle \mathbf{u}, \mathbf{v} \rangle}\end{aligned}$$

Jelikož platí, že $x + \bar{x} = 2\operatorname{Re}(x)$ pro libovolné $x \in \mathbb{C}$, tak potom: $2\langle \mathbf{u}, \mathbf{v} \rangle + 2\overline{\langle \mathbf{u}, \mathbf{v} \rangle} = 4\operatorname{Re}(\langle \mathbf{u}, \mathbf{v} \rangle)$

$$\text{Odkud: } \operatorname{Re}(\langle \mathbf{u}, \mathbf{v} \rangle) = \frac{1}{4}(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2)$$

b)

$$\begin{aligned}\|\mathbf{u} + i\mathbf{v}\|^2 - \|\mathbf{u} - i\mathbf{v}\|^2 &= \langle \mathbf{u} + i\mathbf{v}, \mathbf{u} + i\mathbf{v} \rangle - \langle \mathbf{u} - i\mathbf{v}, \mathbf{u} - i\mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{u} \rangle + \\ &+ \langle \mathbf{u}, -i\mathbf{v} \rangle + \langle -i\mathbf{v}, \mathbf{u} \rangle + \langle -i\mathbf{v}, -i\mathbf{v} \rangle - \langle \mathbf{u}, \mathbf{u} \rangle - \langle \mathbf{u}, -i\mathbf{v} \rangle - \langle -i\mathbf{v}, \mathbf{u} \rangle - \langle -i\mathbf{v}, i\mathbf{v} \rangle = \\ &= -i\langle \mathbf{u}, \mathbf{v} \rangle + i(-i)\langle \mathbf{v}, \mathbf{v} \rangle - i\langle \mathbf{u}, \mathbf{v} \rangle + i\overline{\langle \mathbf{u}, \mathbf{v} \rangle} - i(-i)\langle \mathbf{v}, \mathbf{v} \rangle = i(\overline{\langle \mathbf{u}, \mathbf{v} \rangle} - \langle \mathbf{u}, \mathbf{v} \rangle) + \\ &+ i(\overline{\langle \mathbf{u}, \mathbf{v} \rangle} - \langle \mathbf{u}, \mathbf{v} \rangle)\end{aligned}$$

Jelikož: $i(\bar{x} - x) = 2\operatorname{Im}(x)$ pro libovolné $x \in \mathbb{C}$, tak potom: $2i(\overline{\langle \mathbf{u}, \mathbf{v} \rangle} - \langle \mathbf{u}, \mathbf{v} \rangle) = 4\operatorname{Im}(\langle \mathbf{u}, \mathbf{v} \rangle)$

$$\text{Odtud: } \operatorname{Im}(\langle \mathbf{u}, \mathbf{v} \rangle) = \frac{1}{4}(\|\mathbf{u} - i\mathbf{v}\|^2 - \|\mathbf{u} + i\mathbf{v}\|^2)$$